

```
1 using System;
2 using System.Diagnostics;
3 using System.Globalization;
4 using System.Xml.Linq;
5
6 namespace RestaurantSupplier_National
7 {
8
9     internal class Dairy: Food
10    {
11        private String symbol;
12        private String name;
13        private double price;
14        private int kg;
15        private String type = "Dairy";
16        private double cost;
17
18        public Dairy() : base()
19        {
20
21        }
22
23        public Dairy(String sym, String nam, double pri, int kg)
24        {
25            symbol = sym;
26            name = nam;
27            price = pri;
28            this.kg = kg;
29        }
30
31
32        public override double getValue() //SC13
33        {
34            return price * kg;
35        }
36
37
38        public override double getProfit()
39        {
40            return ((getValue()-5000)/5000);
41        }
42        //SC14
43        public override double DaysSalesOfInventory()
44        {
45
46            return (int)((kg / costOfGoodsSold()) * 365); // Assuming
47                annual sales period
48        }
49        //SC15
```

```
49     public override double costOfGoodsSold()
50     {
51         double tenPercent = kg * 0.1;
52         Random rnd = new Random();
53         cost = (rnd.NextDouble() * (tenPercent - 1) + 1) * 20;
54
55         return cost;
56     }
57     //SC16
58     public override String ToString()
59     {
60         return $"Food Details:\n" +
61             $"- Symbol: {getSKU()}\n" +
62             $"- Food Type (P or D): {type}\n" +
63             $"- Food Name: {getName()}\n" +
64             $"- Price: {getPrice().ToString("C", CultureInfo.CurrentCulture)}\n" +
65             $"- Total Tons: {kg}\n" +
66             $"- Total Value: {getValue().ToString("C", CultureInfo.CurrentCulture)}\n" +
67             $"- Days of Sales Inventory: {DaysSalesOfInventory().ToString()}\n" +
68             $"- COST of Goods Sold: {costOfGoodsSold().ToString("C", CultureInfo.CurrentCulture)}\n";
69     }
70
71
72     //Returns number kg
73     public override int getTons()
74     {
75         return kg;
76     }
77
78     //Returns symbol
79     public override String getSKU()
80     {
81         return symbol;
82     }
83
84     //Returns name
85     public override String getName()
86     {
87         return name;
88     }
89
90     //Returns price
91     public override double getPrice()
92     {
93         return price;
```

```
94     }
95
96     //Adds the number of requested kg
97     public override void procureKG(int s)
98     {
99         kg += s;
100    }
101
102    //Sells the number of requested kg
103    public override void distroKG(int s)
104    {
105        kg -= s;
106    }
107
108
109    }
110 }
```